Array Addressing

Row Major and Column Major Order

Row-major order

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Column-major order

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Address Calculation in 1-D Array

- Memory address calculation Address of an element of an array say A[I] is calculated using the following formula:
  Address of A [I] = B + W * (I- LB) Where,
  - B = Base address
  - W = Storage Size of one element stored in the array (in byte)
  - I = Subscript of the element whose address is to be found
  - LB = Lower limit / Lower Bound of subscript, if not specified, assume 0 (zero)
- Example:
- Given the base address of an array B[1300..1900] as 1020 and the size of each element is 2 bytes in the memory. Find the address of B[1700].
  Ans: The given values are: B = 1020, LB = 1300, W = 2, I = 1700

  Address of A [I] = B + W * (I -LB), each element's size

  = 1020 + 2 * (1700-1300)

  = 1020 + 2 * 400 = 1820

Address Calculation in 2-D Array

- A is a 2D array with dimensions M1 X M2. So that here the total number of elements is equal to M1 X M2. The address should be calculated using one of the two methods:

A. Row-by-Row
   - Location (A [j, k]) = Base (A) + w [N (j-1) + (k -1)]
   - Here, j and k indicate the index value of the element whose address you want to search.
   - Base (A) is the starting address of the given array. W is the memory word required to store given array's element. The array is of size M(rows) X N(columns).

B. Column-by-Column
   - Location (A [j, k]) = Base (A) + w [M (k-1) + (j -1)]
   - Here, j and k indicate the index value of the element whose address you want to search.
   - Base (A) is the starting address of the given array. W is the memory word required to store given array's element. Here, array is the of size M(rows) X N(columns)

Address Calculation in 2-D Array

- Examples
- Consider 30 X 4 2D array and the base address is 200 and 1 word per memory location. Find out the address of A (15, 3). Given Base (A) = 200, M =30, N=4, j=15, k=3, w=1

A. Row-by-row

Location (A [j, k]) = Base (A) + w [N (j-1) + (k -1)]

Location (A [15, 3]) = 200 + 1 [4 (15 - 1) + (3 - 1)]

= 200 + 1 [56 + 2] = 258

B. Column-by-column

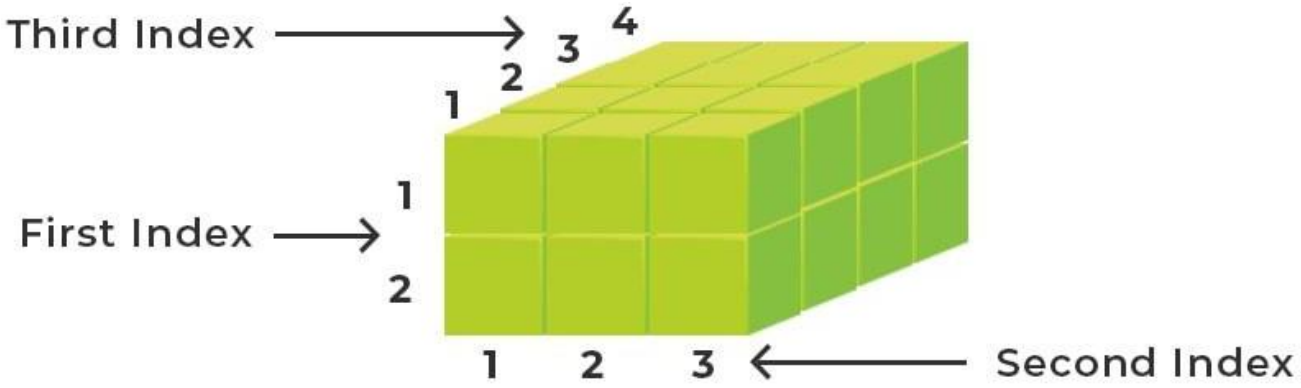
Location (A [j, k]) = Base (A) + w [M (k-1) + (j -1)]

Location (A [15, 3]) = 200 + 1 [30 (3 – 1) + (15 – 1)]

= 200 + 1 [30 (2) + 14] = 274

Address Calculation in 3-D Array

- Suppose A is a 3D array of size M1 X M2 X M3. Total number of elements are M1 X M2 X M3. In 3D array lower bound (LB) is not always zero (0). It can be any number. Base address of any row of matrix is called effective address.



- A [2:11, 3:15] indicates that A is a 2D array with lower bound (LB) of row is 2 and Upper Bound (UB) of row is 11. In the same way lower bound (LB) of column is 3 and upper bound (UB) of column is 15.

Address Calculation in 3-D Array

- Length of each dimension is equal to Li = UB – LB +1
- Lower bound (LB) may not always start with zero (0). So, we have to calculate effective address of each index.
- Effective Address (Ei) = Ki - LB
- Real address of any element is calculated as follows,
- Row-Major: Base (A) + W [(E1L2 + E2) L3 + E3]
- Column-Major: Base (A) + W [(E3L2 + E2) L1 + E1]

Address Calculation in 3-D Array

- Example

  Suppose multidimensional arrays A and B are declared using A [-2:2, 2:22] and B [1:8, -5:5, -10:5].

  Here A is a 2-D array while B is a 3-D array

  1. Find out the length of each dimension and the number of elements in arrays A and B.

  2. Consider B [3, 3, 3] elements in array B, find effective addresses E1, E2 and E3. Also, find the real address of these elements. (Assume the base address of array B is 400, and W is 4)

Address Calculation in 3-D Array

Answer:

For array A

L1 = UB –LB +1

= 2 – (-2) +1 = 5

L2 = UB –LB +1

= 22 – 2 + 1 = 21

For array B

L1 = UB –LB +1

= 8 – 1 + 1

= 8

L2 = UB –LB +1

= 5 – (-5) +1

= 11

L3 = UB –LB +1

= 5 – (-10) +1

= 16

Total number of elements in array A is L1 X L2 = 5 X 21 = 105.

Total number of elements in array B is: L1 X L2 X L3 = 8 X 11 X 16 = 1408.

Address Calculation in 3-D Array

Answer:

Ki = 3, 3, 3 therefore K1 = 3, K2 = 3 and K3 = 3

LB1= 1, LB2 = -5 LB3 = -10

E1 = K1 – LB1 = 3 – 1 = 2

E2 = K2 – LB2 = 3- (-5) = 8

E3 = K3 – LB3 = 3 – (-10) = 13

The real address for Row Major:

= Base (A) + W [(E1L2 + E2) L3 + E3]

= 400 + 4 [(2 (11) + 8) 16 + 13]

= 400 + 1972

= 2372

The real address for Row Major:

= Base (A) + W [(E3L2 +E2) L1 + E1]

= 400 + 4 [(13(11) + 8) 8 + 2]

= 400 + 4840

= 5240